Collaborative Asset Management Using GitHub

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Today's Learning Objectives

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- Understanding of working collaboratively using GitHub
- Creating and cloning repositories
- Understanding common Git terminologies

Pop Quiz

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Q. Which of the following is not a Git hosting service?

a. Bitbucket

b. Gitkraken

c. AWS CodeCommit

d. Codebase



Pop Quiz

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a. Bitbucket

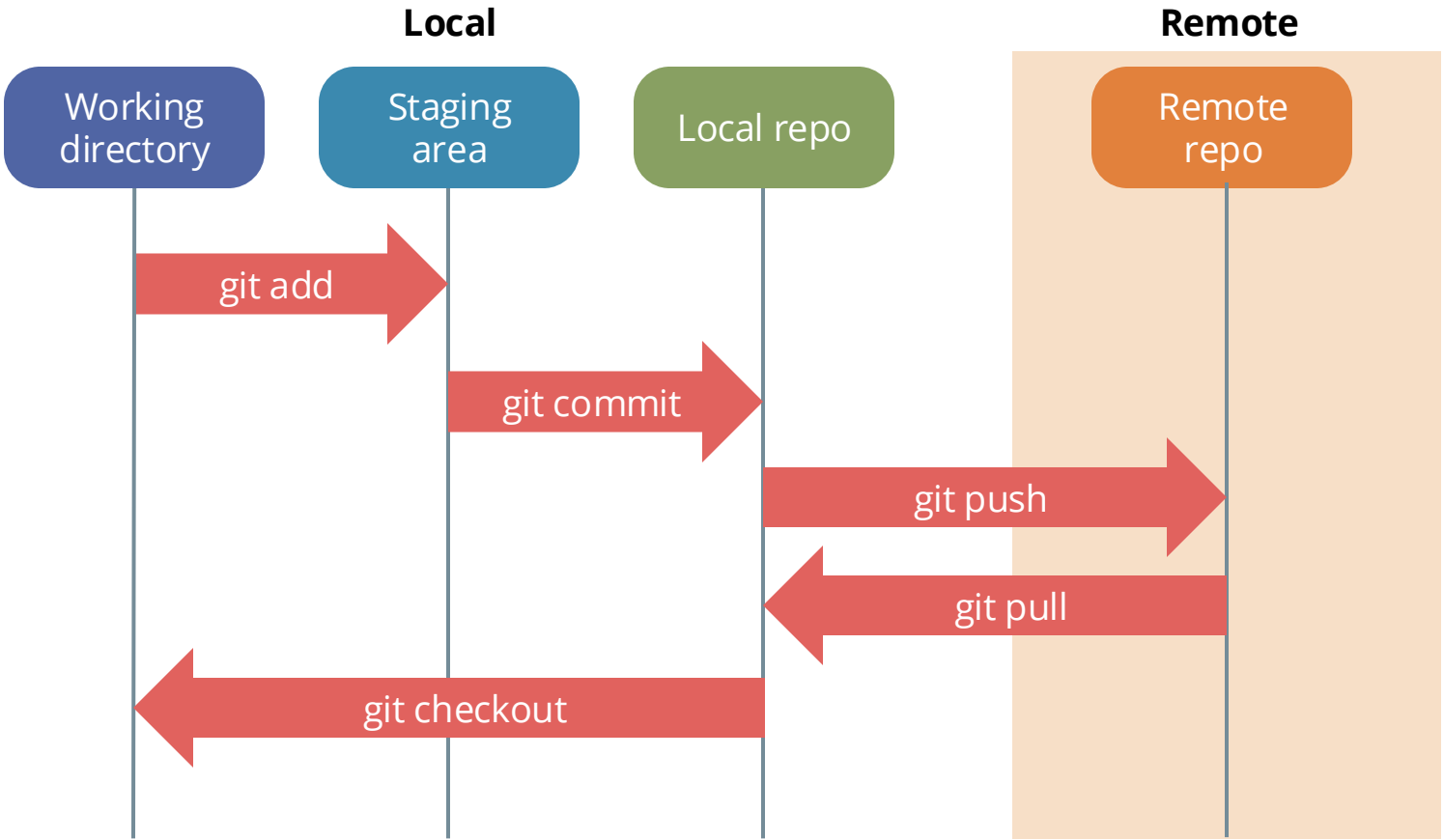
b. Gitkraken

c. AWS CodeCommit

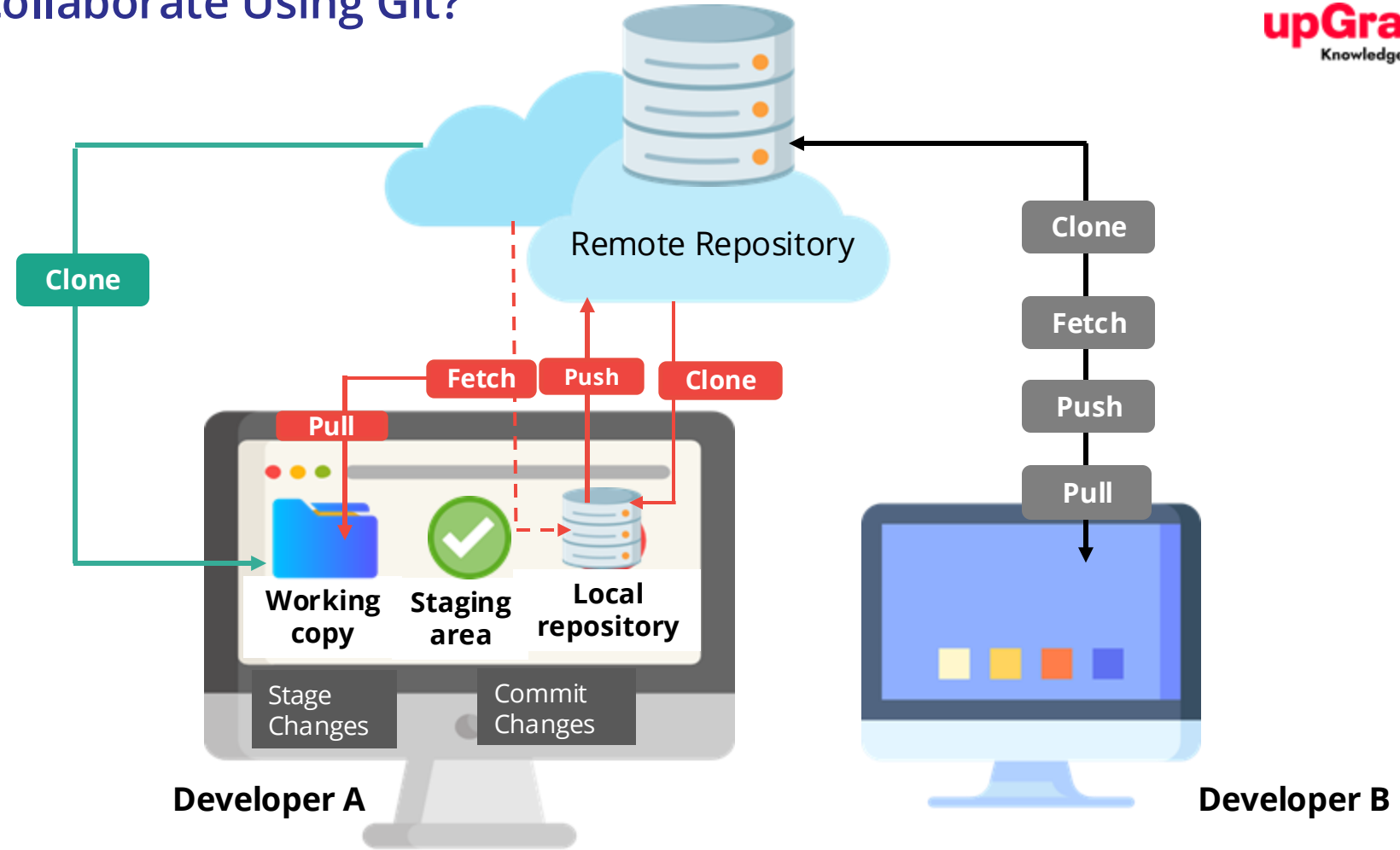
d. Codebase



Git Ecosystem

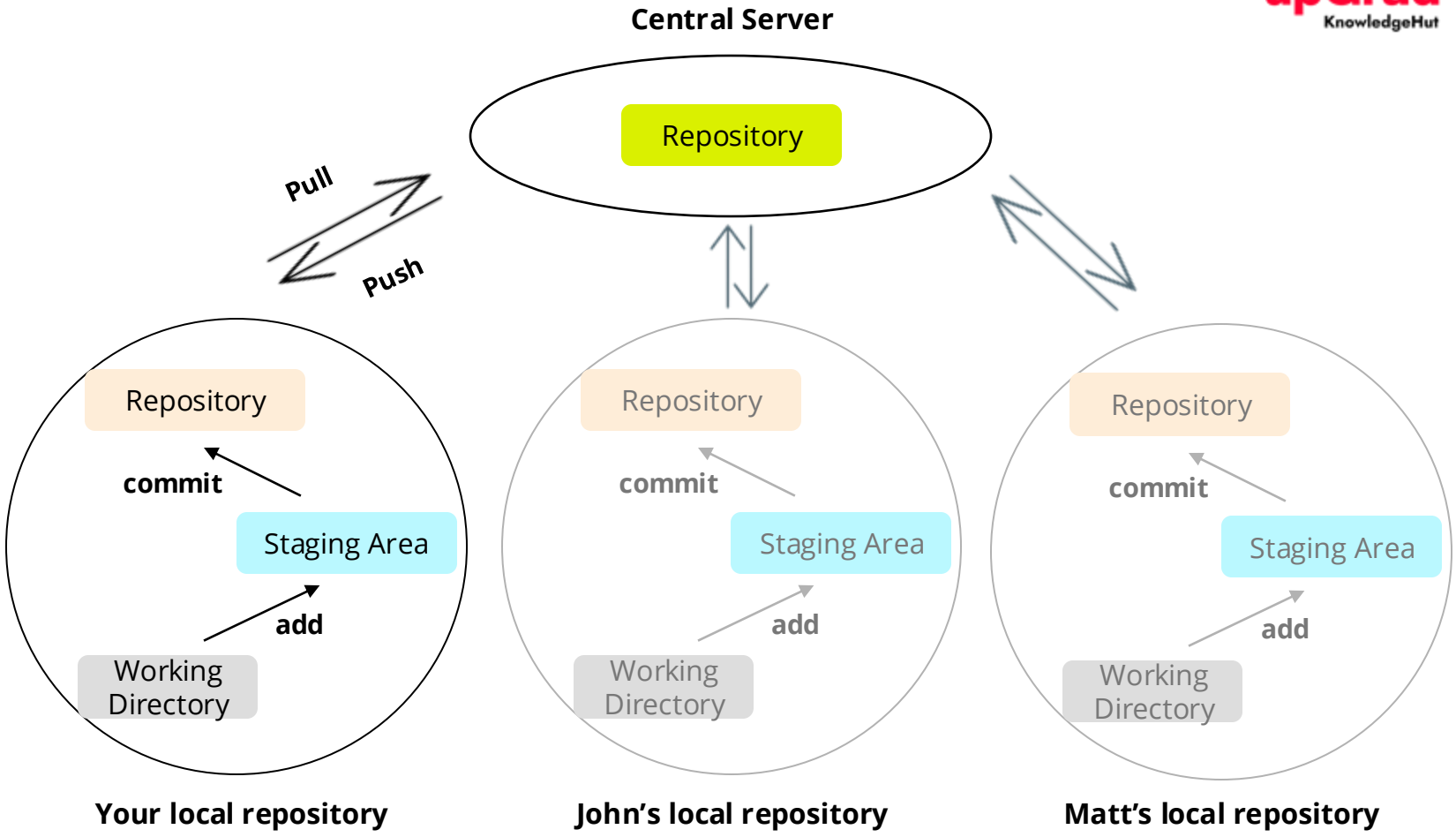


Why Collaborate Using Git?



GitHub for Collaborative Work

Git for Collaboration



GitHub for Collaborative Work

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- GitHub is a collaborative platform.
- The resources can be shared across multiple teams and locations.
- It has powerful version control features.
- Multiple stakeholders can access or edit the same piece of code.
- GitHub is the home for thousands of Open-source Software (OSS) projects.

For installation: http://bit.ly/3RBgTQT

GitHub

Poll Time

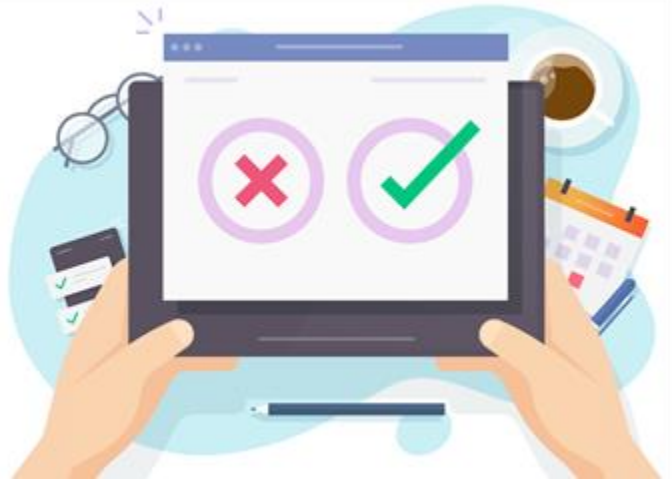
Q. Which of the following describes the central Git repository?

a. Local Repository

b. Remote Repository

c. Main Repository

d. Master Repository



Poll Time

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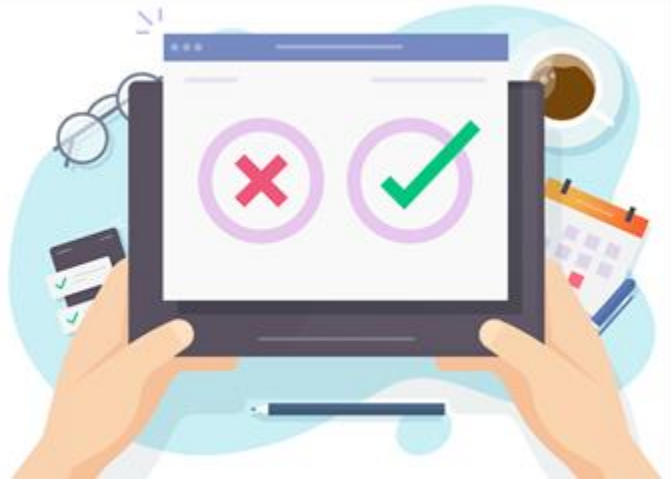
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Git Credential Manager (GCM)

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Git Credential Manager (GCM) ensures the authentication of the Git repositories. It is easy and secure irrespective of where and how the code is stored.

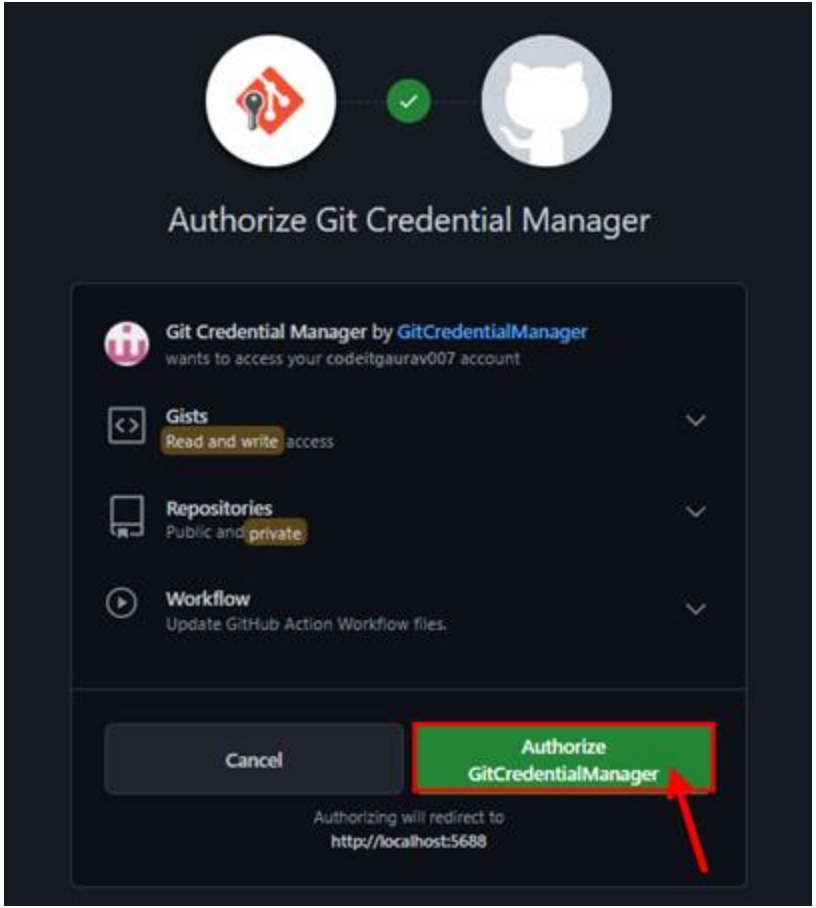


Installation of GCM:

http://bit.ly/3DMvRhl

Git Credential Manager (GCM)

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Git Credential Manager (GCM)

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microsoft/**Git-Credential-Manager-for-Windows**



Secure Git credential storage for Windows with support for Visual Studio Team Services, GitHub, and Bitbucket multi-factor authentication.

52	76	3k	646
Contributors	Issues	Stars	Forks



Pop Quiz

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Q. What is the purpose of Git Credential Manager?

a. To eliminate the need for users to enter their Git credentials repeatedly

b. To manage Git repositories

c. To automate the Git workflow

d. To improve the performance of Git operations



Pop Quiz

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GitHub Portfolio

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- For developers, creating repositories on GitHub is the best way to create a portfolio to showcase skills.

- Collaborators can be easily added to projects so that they can contribute to the repository.

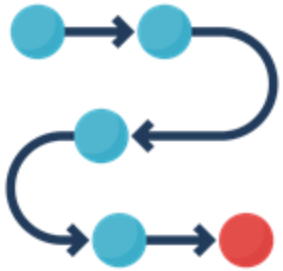


GitHub: Initial Workflow

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Steps to use GitHub

1. Go to the link: https://github.com/.
2. Click on “Sign up for GitHub”. Fill up and submit the form to create your free account.
3. Click on “Start a new project”.
4. Enter any repository name and click on “Create Repository”. Give a description to your repository.
5. Once the repository is created, you are ready to commit, pull, push and perform all the other operations.
6. Create branches and perform operations: commit, pull, push, merge, etc.



Adding Collaborators

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To add a collaborator, go to the repository in GitHub. Then to **Settings -> Manage Access, Click on Invite Collaborators,** and type the username or email address to send an invitation.



Pop Quiz

Q. Why should you put your work on GitHub?

a. To showcase your personal projects

a. To take inputs from the community of developers

a. To maintain your portfolio for prospective employment opportunities

a. All the above



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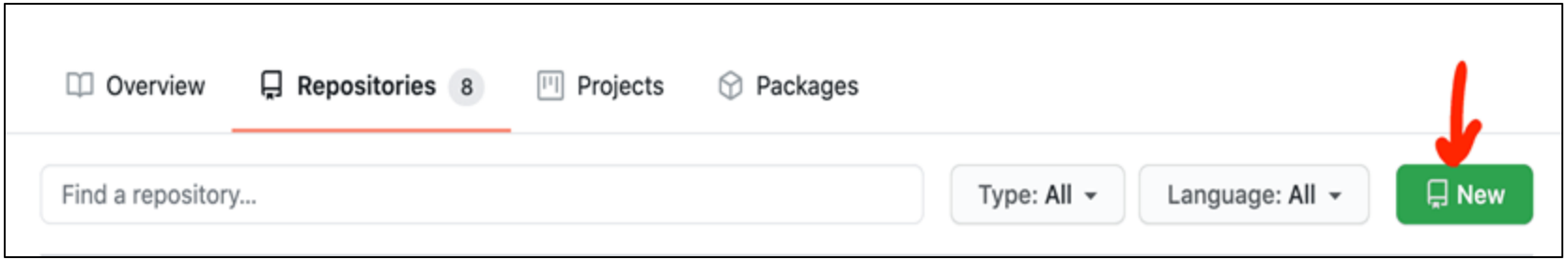
a. All the above



Create a GitHub Repository

Steps to create a GitHub repository:

1. Open your GitHub profile (github.com/<your_username>).
2. Go to the 'Repositories' tab.
3. Click the 'New' button on the upper-right corner, and from the dropdown menu, select "New Repository".
4. Create a new repository by clicking the 'New' button.



Create a GitHub Repository – Activity

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1. Log in to your GitHub account or sign up for one if you don't have one already.
2. Once logged in, click on the "+" icon on the top right corner of the GitHub dashboard and select "New repository".
3. Give your repository a name and a description. Depending on your needs, you can make it public or private.
4. If you want to add a README file, click on the checkbox that says, "Add a README file".
5. Choose a license for your repository. A license determines how others can use your code. If you're unsure which license to choose, leave it as "None" for now.
6. Click on the "Create repository" button at the bottom of the page.
7. Congratulations! You've just created your first GitHub repository. You can now add files, make changes, and collaborate with others on your project.

Common Terminologies

Frequently Used Terms by GitHub Users

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- **Creating a repository:** It is for collaborative work.

- **Main branch in a repository:** This is the final version that is considered ready to use by the team or outside if the repository is public. This was previously known as the 'master' branch.

- **Creating a branch:** To try new ideas, create branches in your project. You can make changes in the branch without affecting the master unless a pull request is accepted.

- **Adding commits:** It is to keep track of the progress on the branch.



Frequently Used Terms by GitHub Users

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- **Forking a repository:** It creates a copy for you to work on independently without any changes to the others. If needed, send a pull request to the owner so that the owner can incorporate the recent amendments.

- **Pull requests:** These initiate a discussion about your commits or changes made to a code.

- **Issues:** It highlights bugs or issues with codes that need correction. Issues remain open unless resolved, which can be filtered out and labeled as a bug.

- **Markdown:** It is a way to style text on the web.



Poll Time

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Q. How do you fork a GitHub remote repository?

a. git fork <url_of_remote_repo>

b. Create a new repo and use git clone

c. Using the UI interface

d. Create a new repo and use git pull



Poll Time

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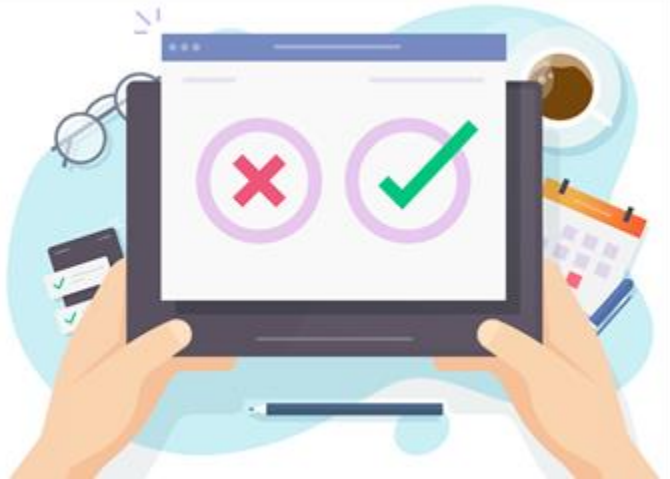
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Cloning

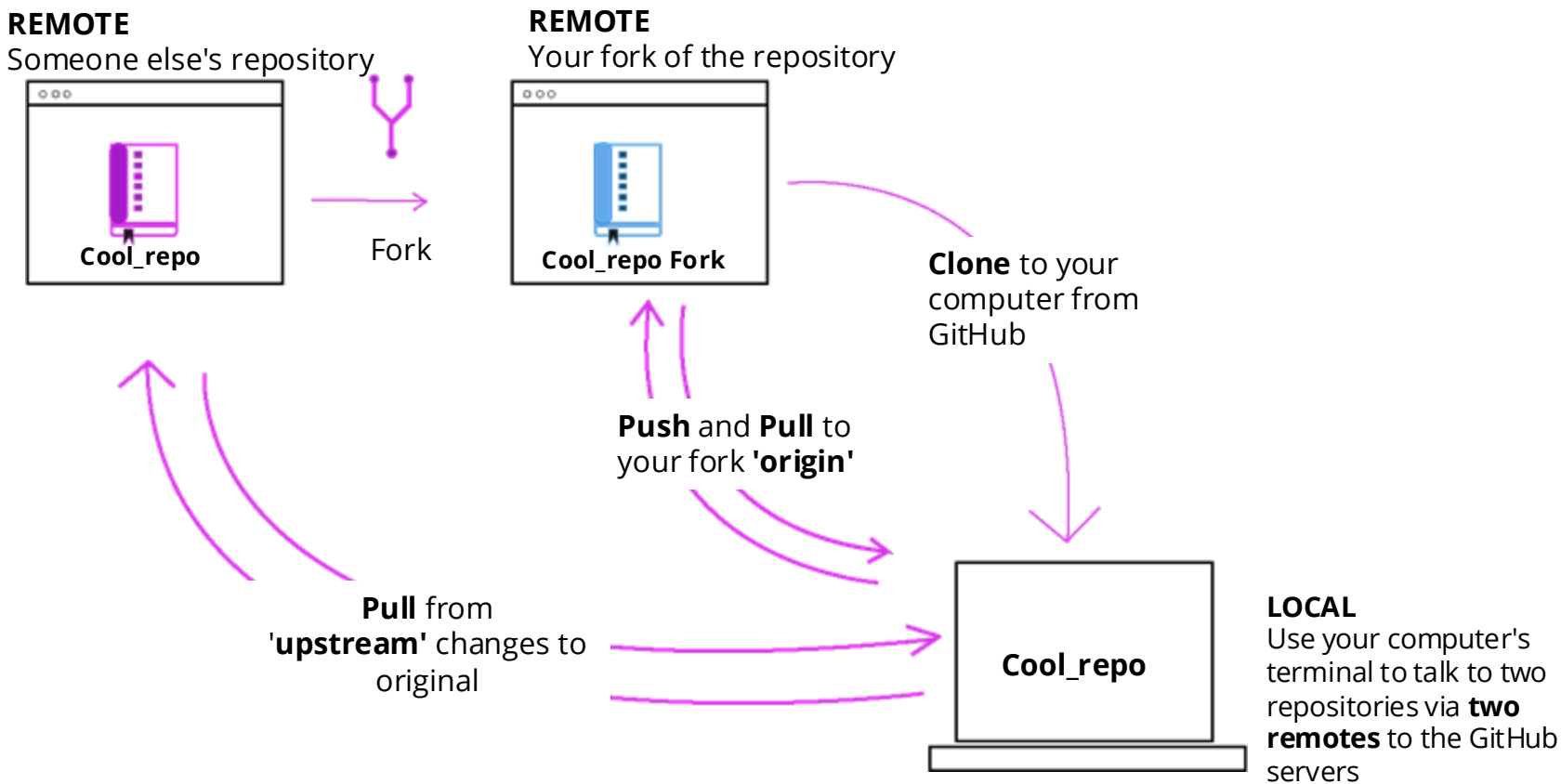
Cloning

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- Create an identical local copy of your work based on the online copy in the remote repository.
- When you clone a repository, all the files are downloaded to the local machine, but the remote git repository remains unchanged.
- You clone a repo to work locally and push changes to the online repository.
- Enables multiple developers to work on the same project by contributing to the online repository.



Need for Cloning



Poll Time

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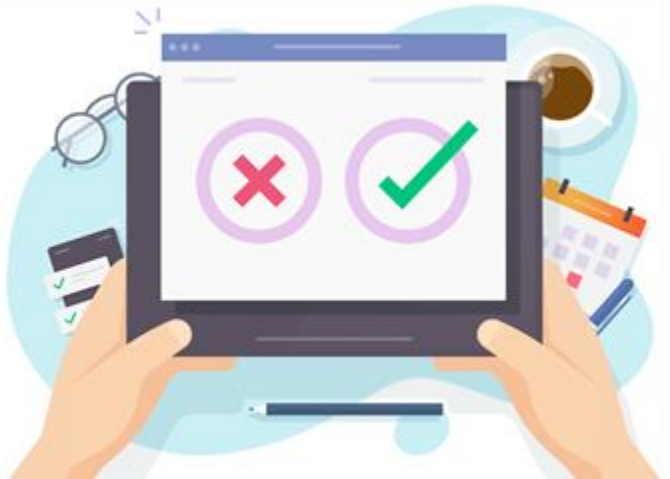
Q. What is the syntax for cloning a Git repository?

a. git clone [repository URL]

b. git clone [repository name]

c. git clone [username/repository name]

d. All the above



Poll Time

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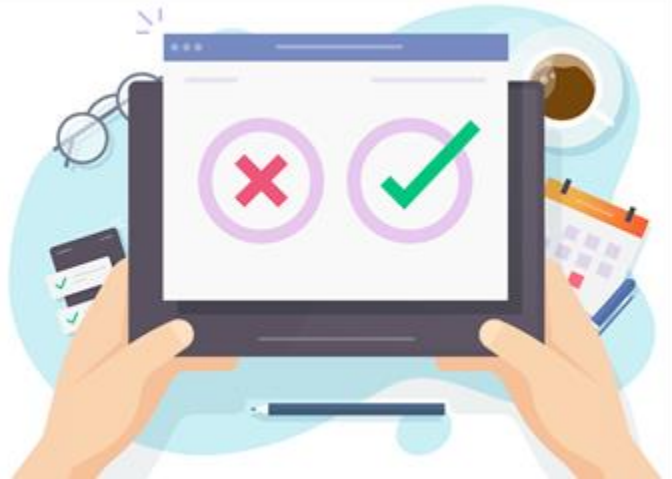
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Clone a Repository

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1. The user starts from the upstream repository on GitHub and clones the repository into their local machine.

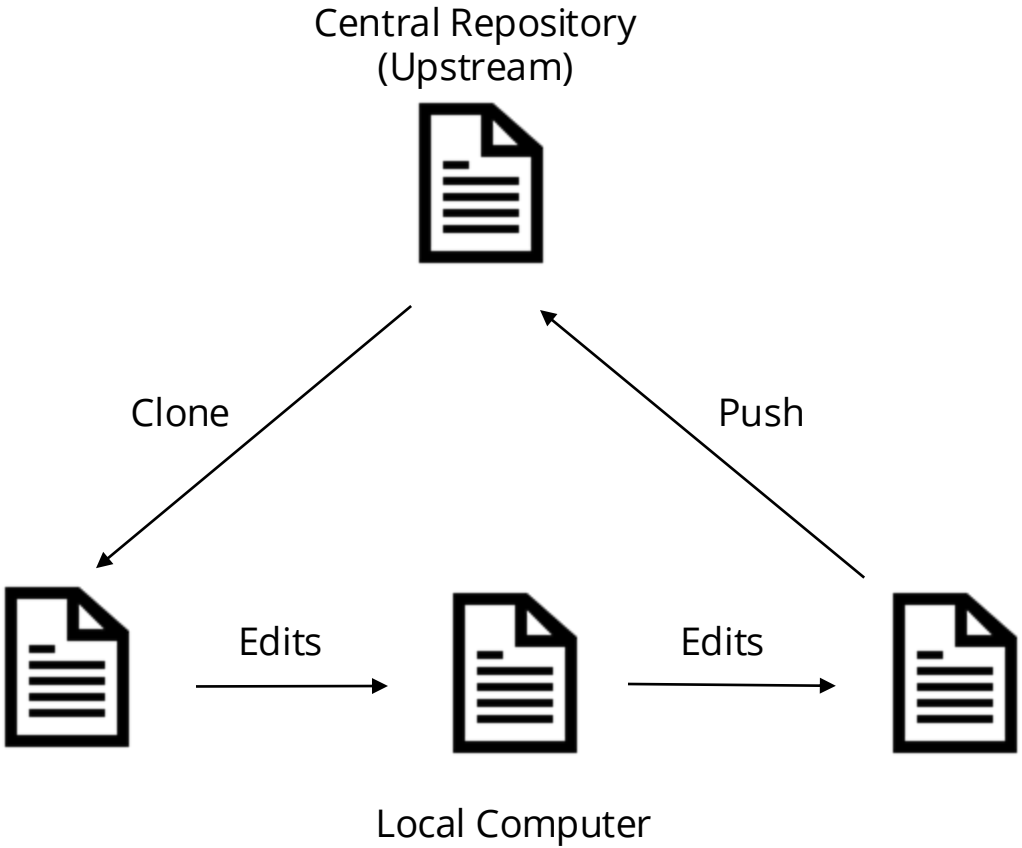
1. After cloning, contributors provide their contribution to the local repository.

1. Iteration, testing, and refinement.

1. Once the changes are done, all the modifications can be pushed to the upstream repository.



Cloning


Central Repository
(Upstream)
Clone
Push
Edits
Edits
Local Computer

Poll Time

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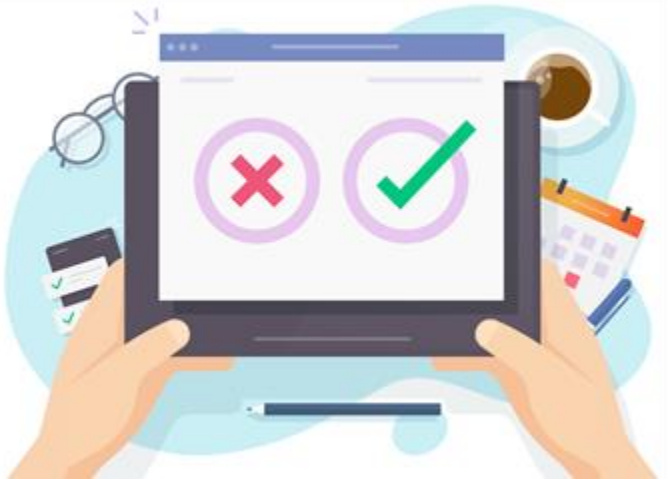
Q. How do you push a local branch 'feature/notifications' along with the code to the remote repo that doesn't have this branch?

a. git push -u

b. git push --set-upstream 'feature/notifications'

c. git push --set-upstream origin 'feature/notifications'

d. git push –u 'feature/notifications'



Poll Time

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Activity

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1. Create a repository on GitHub called 'git-tutorial'.
1. Clone the repository locally.
1. Create a branch called '**feature/jedi-master**'.
1. Create a file 'StarWars.txt' with the content 'a long time ago, in a Galaxy far far away'.
1. Create another file called **.env** with the content '**DB_URL=http:localhost:20457**'.
1. Add a folder called **node_modules.**
1. Add **.env** file and **node_modules** folder to **.gitignore.**
1. Commit the files to the current branch ie. '**feature/jedi-master**'.
1. Push the branch along with the files to the remote repo.

Summary

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- GitHub is a cloud hosting service for codebases. It is used by enterprises, open-source organizations, and individual contributors alike.
- GCM is an easy way to manage credentials across different repos.
- GitHub provides a UI where we can create repository, add branches, commit code, merge branches, create pull requests among other things.
- Forking creates a copy of the original repository.
- We can add collaborators to a repository so we can work in teams.
- **git push –u/--set-upstream origin <branch_name>** allows you to push a branch along with its code.
- **git config –global push.autoSetupRemote true** will enable auto tracking to upstream branches. If the branch doesn't exist on remote, it will be created and tracking will be enabled.

Next Session:
Advanced GitHub Functions

Thank You

Please remember to complete
your assessments and learning for
the week on
PRISM

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